

Ensemble Methods: Boosting vs. Bagging

From-scratch Decision Trees, AdaBoost & Random Forests

Gurban Burjuzada Murad Asgarov Davud Gurbanov

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Team StarPlatinum

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Motivation

What We Built

Experimental Setup

Results: Scaling Behaviour

Results: Head-to-Head

Results: Robustness

Unsupervised Analysis

Discussion & Conclusion

“Under what conditions does boosting outperform bagging, and vice versa — and why?”

Boosting (AdaBoost)

- Sequential, reweights hard examples
- Additive stumps, $\alpha_m = \ln \frac{1-\epsilon_m}{\epsilon_m}$
- Reduces bias
- Sensitive to label noise

Bagging (Random Forest)

- Parallel, independent bootstrap trees
- Feature sub-sampling decorrelates trees
- Reduces variance
- Robust to noise, multi-class

Our approach

Implement all algorithms **from scratch** (NumPy only) and answer the question empirically across four datasets of varying size, dimensionality, and balance.

Supervised

- **Decision Tree** — CART, exhaustive split search, Gini/entropy, sample-weight support, feature importances
- **AdaBoost (SAMME)** — depth-1 stumps, staged predict, multi-class α_m correction
- **Random Forest** — bootstrap + $\sqrt{p}$ feature sub-sampling, OOB score, parallel via multiprocessing

Unsupervised

- **PCA** — covariance eigen-decomposition
- **K-Means** — Lloyd's algorithm, empty-cluster re-init
- **DBSCAN** — brute-force region queries, k -distance heuristic for ε

Baseline sanity check

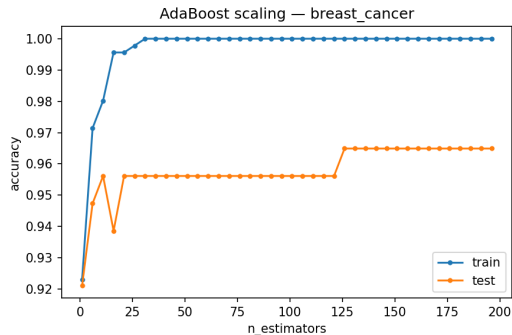
Our tree matches sklearn within **0.3%** accuracy on all datasets (tolerance: 2%).

Dataset	Samples	Feat.	Cls	Challenge
Breast Cancer	569	30	2	clean baseline
Adult Income	48 842	14	2	imbalance ($\leq 1\%$)
Covertypes (subset)	5 000	54	7	multi-class
MNIST Binary (0/1)	$\sim 6\,000$	784	2	high-dim

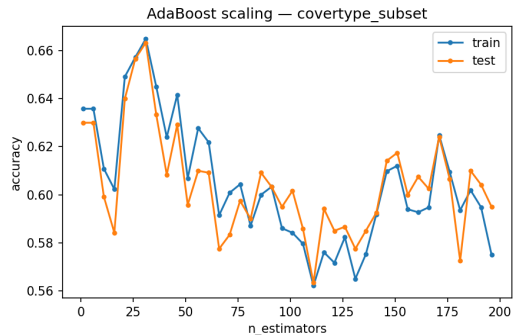
Protocol: standardised features, 5-fold CV, seed 42, metrics = accuracy / macro-F1 / AUC-ROC.

7 experiments: baseline vs. sklearn, AdaBoost & RF scaling, head-to-head CV, noise robustness, bias–variance, unsupervised analysis.

AdaBoost: Clean Data vs. Multi-Class Data



Breast Cancer: saturates ~50 stumps, **no overfitting**

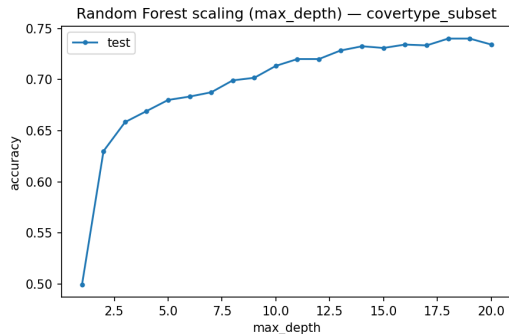
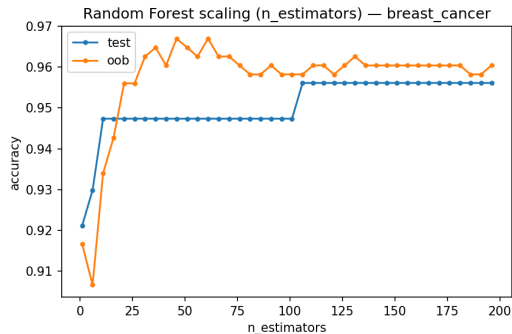


Covertype: test accuracy **peaks then degrades** while train $\rightarrow$ 0.95

Key insight

On clean binary data AdaBoost's margin-maximising objective keeps improving; on multi-class noisy data, sequential reweighting starts chasing mislabelled points $\Rightarrow$ overfitting.

Random Forest: Trees and Depth



Breast Cancer: test & **OOB** accuracy track closely, saturate ~50 trees
Coverytype: deeper trees matter — 0.499 (depth 1) → 0.740 (depth 18)

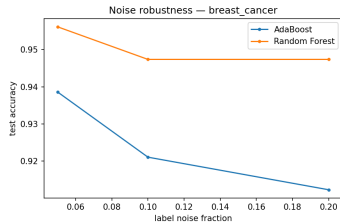
- OOB error is a reliable *free* internal estimate of test error
- Unlike AdaBoost, RF shows **no overfitting** as capacity grows

5-Fold CV Accuracy (100 estimators)

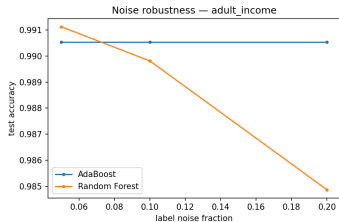
Dataset	Tree	AdaBoost	RF (ours)	RF (sklearn)
Breast Cancer	0.912	0.977	0.961	0.966
Adult Income	0.983	0.991	0.991	0.990
Covertypes	0.686	0.622	0.760	0.758
MNIST Binary	0.996	0.999	0.999	0.999

- AdaBoost wins on **clean** data (Breast Cancer, MNIST)
- RF wins by **+14 pts** on **multi-class** Covertypes (0.760 vs. 0.622)
- Our RF matches sklearn within 0.5% everywhere
- Macro-F1 tells the **same story** (see report, Table II)

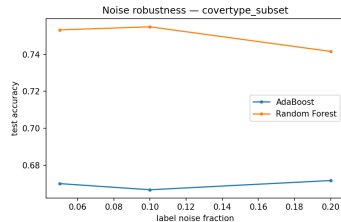
Noise Robustness



Breast Cancer



Adult Income



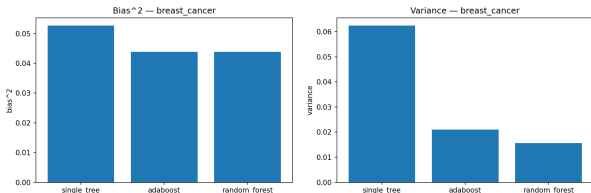
Covertime

- Breast Cancer: AdaBoost drops **0.939** $\rightarrow$ **0.912** at 20% noise; RF degrades more gracefully
- Adult Income: both stable (imbalance masks noise on the majority class)
- Covertime: RF beats AdaBoost by **>10 pts** at every noise level

Bias–Variance Decomposition

Model	Bias ²	Var	Err
Single Tree	0.053	0.062	0.081
AdaBoost	0.044	0.021	0.046
Random Forest	0.044	0.016	0.052

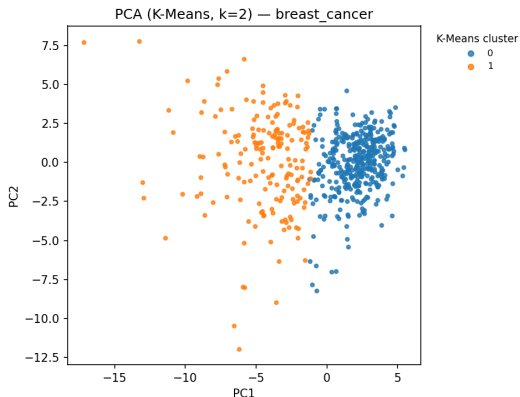
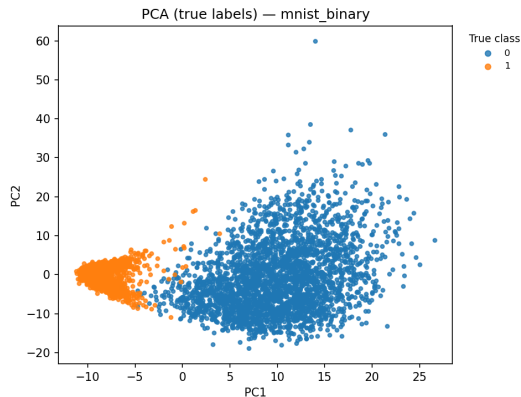
Breast Cancer, 100 bootstrap replicates, 50 estimators.



Confirms theory exactly

Both ensembles cut bias vs. a single tree; **AdaBoost** keeps the lowest average error here, **RF** achieves the lowest variance via bootstrapped averaging.

PCA-Projected Structure: Easy vs. Hard Case



MNIST 0-vs-1: **well-separated** — K-Means ARI = **0.958** Breast Cancer: **poor cluster-class alignment** — ARI = 0.010

Dataset	PCs (90%)	K-Means ARI	DBSCAN ARI
Breast Cancer	6	0.010	0.003
Adult Income	8	0.001	0.001
Coverttype	21	0.020	0.005

Boosting wins when ...

- Data is **clean**, low-noise
- Problem is **binary** / well-separated
- Bias reduction matters most

Bagging wins when ...

- Data is **noisy** or mislabeled
- Problem is **multi-class** / high-dimensional
- Variance reduction matters most

Persistent challenge: class imbalance

On Adult Income both methods reach >0.99 accuracy but only ~ 0.63 macro-F1 — accuracy hides poor minority-class recall. Class weighting alone is insufficient for 1% imbalance.

Takeaways

- Empirical results **match ESL theory**: boosting $\downarrow$ bias, bagging $\downarrow$ variance
- RF is the safer default under noise / multi-class / imbalance
- AdaBoost is hard to beat on clean, well-separated binary problems

Limitations & Future Work

- Stumps only for AdaBoost $\rightarrow$ try deeper base learners
- SAMME.R, gradient boosting (XGBoost)
- Better imbalance handling (SMOTE, cost-sensitive)
- ϵ selection via knee-finding, not percentile

All code, tests, figures and results are fully reproducible.

Thank you!

Questions?